# PG No. 39 $S_4(c)$ $\bar{4}$ [ tetragonal ]

* Wyckoff site: `2a`, site symmetry: `2..`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $[0, 0, z]$ | `[1,-2]` |
| 2 | $[Y, -X, 0]$ | $[0, 0, -z]$ | `[3,-4]` |

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $[0, 0, z]$ | `[1,2]` |
| 2 | $[0, 0, -Z]$ | $[0, 0, -z]$ | `[3,4]` |

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[0, 0, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[0, 0, z]$ | `[2]` |
| 3 | $[Y, -X, -Z]$ | $[0, 0, -z]$ | `[3]` |
| 4 | $[-Y, X, -Z]$ | $[0, 0, -z]$ | `[4]` |

* Wyckoff site: `4b`, site symmetry: `1`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, -y, z]$ | `[2]` |
| 3 | $[Y, -X, -Z]$ | $[y, -x, -z]$ | `[3]` |
| 4 | $[-Y, X, -Z]$ | $[-y, x, -z]$ | `[4]` |